MLFF Architecture Revision 45

PERF-P0 exact native TARGET-DATA2B closure

mdstats project

2026-08-15

Scope

Revision 45 closes bounded `PERF-P0`, the first implementation gate after the `PERF-BASE0` oracle. It advances TARGET-DATA2B storage and exact execution only. The coverage policy, target membership, FP64 authority, fixed-mass rule, leave-one-out convention, tie behavior, and downstream scientific decisions are unchanged.

The authoritative release is `mdstats 0.20.179a0`. `PERF-P1` is next.

Implemented authority

Native scientific arrays

TARGET-DATA2B family references now own canonical little-endian, C-contiguous, read-only NumPy arrays:

```
frame_indices : <i8 [N]
values        : <f8 [N,D]
weights       : <f8 [N]
scales        : <f8 [D]
local_radii   : <f8 [N]
```

Every array is authenticated by dtype, shape, byte count, and streamed SHA-256 of canonical native bytes. SHA-256 follows FIPS 180-4 [3].

Native persistence

`mdstats.training_data.target_coverage_store` writes content-addressed NPY shards with an authenticated manifest and campaign pointer. NPY was selected because it preserves dtype and shape and supports memory mapping without expanding arrays through Python scalar objects [1,2].

The store:

- deduplicates identical frame-index/weight profiles across families;
- writes through a temporary directory and atomically promotes the completed record;
- validates pointer, manifest, relative path, file size, file SHA-256, dtype, shape, and array-byte identity on read;
- restores large arrays through read-only mmap above a configured threshold; and
- excludes mmap paths, thresholds, query block sizes, worker counts, timing, and host observations from scientific identity.

Historical inline `mdstats.target-coverage-reference.v1` remains readable. It is promoted to v2 only after exact elementwise comparison produces a passing `TargetCoverageMigrationReport`.

Exact execution realization

The scientific definition is retained while removing redundant work:

1. Correlation-unit-balanced frame weights are cached by exact domain, membership, unit, weighting, and leave-one-out identity.
2. Each scalar column is stably ordered once; all required weighted quantiles are derived from the same cumulative mass. Quantile conventions are stated explicitly because software packages do not share one universal definition [4].
3. Uniform weights use an exact fixed-rank neighbor dispatch. The historical cumulative-weight implementation remains the oracle for duplicates and ties.
4. Profile families expose columnar value/missing arrays rather than repeated Python tuple construction.

5. A bounded dense exact kernel is retained for qualification. SciPy `cKDTree` remains production authority; no approximate-neighbor path is introduced [5,6].

For a uniform valid population of size N, leave-one-out mass $M_i = 1 - 1/N$, and target fraction β, the required neighbor rank is

$$k = \lceil \beta(N-1) \rceil.$$

This is an execution dispatch for the frozen cumulative-mass rule, not a new coverage definition.

Supplied-data qualification

The exact benchmark uses all 27 supplied target XML sources:

Quantity	Authority
Target frames	37,633
Target atoms	6,322,344
Coverage families	8
Family elements	263,398
Numerical arrays compared	48

All 48 array identities match the PERF-BASE0 `target_data2b_exact_radii` stage. All five historical-path and five PERF-P0 runs share scientific digest

`2f04aba96b876a10e62b7b0c22f26b544836005d11bc893177d4b29cbe4a3f82`.

Exact construction

Path	Median wall	Observed range	Median process CPU	Median peak RSS
Pre-P0	7.541 s	6.826–9.926 s	27.091 s	329.47 MiB
PERF-P0	6.236 s	5.818–8.253 s	26.043 s	328.50 MiB

The matched median wall reduction is

$$100\left(1 - \frac{6.236470132}{7.541068095}\right) = 17.30\%.$$

The full observed range is retained because host scheduling noise is material.

Persistence

At complete supplied-data scale, native v2 is 56.22x faster to write, 79.70x faster to read, and 58.10% smaller. Both restored references have content digest

`4f46dfaa6c366ede1cd4d19cf5fb8be65cfe49067b5aff85d668e0078915f9b8`.

The migration report contains no difference paths and has digest

`bbe21f1c20beaefb7c837ec20330365853727366f07e4b8a795745aa048bfd88`.

Evidence and limits

Normative and measured evidence is stored in:

- `docs/history/mlff/retired_specs/mlff_perf_p0_native_target_coverage_spec.{md,pdf}`;
- `audits/analysis/mlff_perf_p0_lta_cloud_cpu_2026-08-15.{json,md,pdf}`; and
- `release/MLFF_PERF_P0_QUALIFICATION_0.20.179a0.json`.

No authorizing MACE-MH-1 checkpoint, GPU runtime, or complete production campaign bundle was supplied. This revision does not claim production DATA6 model-derived families, complete TARGET-DATA2C/DATA7 authority, DATA8 materialization, TRAIN2/EVAL2 timing, GPU memory, or OOM evidence.

`PERF-P1` must reuse exact FPS state and progressive coverage state across TARGET-DATA2C and DATA7 while preserving the PERF-BASE0/P0 oracle.

References

[1] R. Kern, “A Simple File Format for NumPy Arrays,” NumPy Enhancement Proposal 1, 2007. Available at: https://numpy.org/doc/1.13/neps/npy-format.html (accessed 2026-08-15).

[2] NumPy developers, “numpy.load,” NumPy reference documentation. Available at: https://numpy.org/doc/stable/reference/generated/numpy.load.html (accessed 2026-08-15).

[3] National Institute of Standards and Technology, *Secure Hash Standard (SHS)*, FIPS PUB 180-4, 2015. DOI: 10.6028/NIST.FIPS.180-4.

[4] R. J. Hyndman and Y. Fan, “Sample Quantiles in Statistical Packages,” *The American Statistician* **50**(4), 361–365 (1996). DOI: 10.1080/00031305.1996.10473566.

[5] J. L. Bentley, “Multidimensional Binary Search Trees Used for Associative Searching,” *Communications of the ACM* **18**(9), 509–517 (1975). DOI: 10.1145/361002.361007.

[6] SciPy developers, “scipy.spatial.cKDTree,” SciPy reference documentation. Available at: https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.cKDTree.html (accessed 2026-08-15).